

A pseudo-spectral algorithm for the computation of transitional-mode eigenfunctions in loose transition states

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Abstract

An algorithm is presented for the exact action of the Hamiltonian describing the interfragment motions in a loose transition state. The working basis set consists of body-fixed Wigner functions, in terms of which the rotational kinetic energy operator is almost diagonal. The action of the potential operator is effected by transformation to a discrete variable representation. Eigenfunctions and eigenvalues corresponding to the exact adiabatic channel potentials can be obtained using the Lanczos algorithm. Illustrative calculations are presented for the dissociation of ketene into singlet methylene and carbon monoxide fragments. The algorithm is tailored to deal with very large basis sets.

1. Introduction

The problem of predicting product state distributions for unimolecular dissociation reactions proceeding on a potential surface where there is essentially no barrier to the recombination process has become an intriguing one in recent years with the advent of increasingly precise measurements on certain benchmark reactions. Earlier measurements [1–4] indicated that these reactions appeared to distribute the available energy statistically amongst the rotational degrees of freedom, also termed the transitional modes (TM), whilst the vibrational degrees of freedom of the product fragments carried more energy than the statistical phase space theory would predict. Marcus and Klippenstein [5–7] successfully modeled these results by assuming adiabaticity of the product fragment vibrations after the transition state. Recent measurements, however, have added complexity to the picture. Reisler and co-workers [8] have shown that, for the case of NO_2 dissociation,

although the product state distributions are broadly statistical, detailed measurements reveal sharp structure which can be associated with the quantum-ergodic character of the excited NO_2 resonances [9]. Moore and co-workers have carried out measurements of the product state distributions of methylene arising from the dissociation of ketene in its ground singlet electronic state and find that the CH_2 fragment is rotationally cold [10]. The inability of earlier statistical models to deal with the complexity of dynamics which is being revealed prompts the development of more rigorous dynamical calculations.

In this Letter, we present an efficient algorithm for effecting the exact action of the TM Hamiltonian H_{TM} (i.e. that obtained by treating the molecular fragments as rigid rotors with an appropriately averaged geometry [11]). The method facilitates the action of H_{TM} on a state vector without the need to explicitly store the Hamiltonian matrix, and so can be used efficiently with very large basis sets. As an illustration of the use of the algorithm within an

iterative scheme, we apply the Lanczos algorithm to calculate the multidimensional TM eigenfunctions and energy eigenvalues associated with the tumbling motion of $^1\text{CH}_2$ and CO during the dissociation of ketene in its ground singlet electronic state, a process for which there is no barrier to the reverse recombination reaction. This provides a method for computing exact adiabatic channel potentials (or, correlation diagrams) (e.g. [12–20]). The method also provides a computational kernel for the development of fully quantum algorithms for studying post-transition-state dynamics in reactions where an adiabatic separation of hindered rotational motion from internal fragment vibrations is justified.

2. Theory

Our formulation of the Hamiltonian for the system consisting of two dissociating or recombining fragments stems from the work of Brocks et al. [21]. Based on the evidence of adiabaticity of the fragment internal vibrations between the variationally determined transition state and infinite separation [7], we adiabatically separate the internal fragment vibrations from the radial and tumbling motion of the fragments. For the purposes of studying the dynamics of the latter modes, the fragments may be treated as rigid rotors with rotational constants evaluated for a geometry averaged over the internal vibration state, or even more simply rotational constants evaluated at the equilibrium fragment geometries. It is assumed that the radial separation R between the centers of mass of the two fragments is an acceptable choice of reaction coordinate. In the spirit of statistical theories such as the variational Rice–Ramsperger–Kassel–Marcus (VRRKM) theory [11] and the statistical adiabatic channel model (SACM) [13], we focus on the action of the TM Hamiltonian; hence the radial motion is not explicitly included in the calculations. The TM Hamiltonian is written as the sum of the rotational kinetic energy for each fragment $\hat{K}_F^{\text{ROT}}$ ($F=A, B$), the rotational kinetic energy for the orbital rotation $\hat{K}_{\text{ORB}}$, and an interaction potential $\hat{V}_{\text{INT}}$

$$\hat{H}_{\text{TM}} = \hat{K}_A^{\text{ROT}} + \hat{K}_B^{\text{ROT}} + \hat{K}_{\text{ORB}} + \hat{V}_{\text{INT}}. \quad (1)$$

In the transformation to body-fixed angular coordinates with two-angle embedding, Brocks et al. [21] showed that $\hat{K}_F^{\text{ROT}}$ retains the standard form

$$\begin{aligned} \hat{K}_F^{\text{ROT}} &= A_F (\hat{j}_a^F)^2 + B_F (\hat{j}_b^F)^2 + C_F (\hat{j}_c^F)^2 \\ &= \frac{1}{2} \left[(A_F + B_F) (\hat{j}^F)^2 \right. \\ &\quad \left. + (2C_F - A_F - B_F) (\hat{j}_c^F)^2 \right. \\ &\quad \left. + \frac{1}{2} (A_F - B_F) \left[(\hat{j}_+^F)^2 + (\hat{j}_-^F)^2 \right] \right], \end{aligned} \quad (2)$$

whereas $\hat{K}_{\text{ORB}}$ takes the form

$$\hat{K}_{\text{ORB}} = \frac{\hbar^2}{2\mu R^2} \left(\hat{J}^2 + \hat{j}^2 - 2\hat{j}_z \hat{J}_z - \hat{j}_+ \hat{J}_+ - \hat{j}_- \hat{J}_- \right). \quad (3)$$

In Eq. (2), the rotational constants (with units of energy) and angular momentum operators have their usual meanings. $\hat{J}$ and $\hat{j}$ in Eq. (3) represent respectively the magnitude of the total angular momentum vector $\hat{J}$ and the magnitude of the sum of the two fragment angular momentum vectors $\hat{j} = \hat{j}_A + \hat{j}_B$, so that

$$\hat{J} = \hat{j} + \hat{l} = \hat{j}_A + \hat{j}_B + \hat{l}. \quad (4)$$

$\hat{j}_z$ and $\hat{j}_\pm$ are the corresponding body-fixed z -axis projections, and $\hat{J}_\pm$ and $\hat{j}_\pm$ the raising and lowering operators.

We have chosen as a working representation a direct-product basis set consisting of body-fixed Wigner functions,

$$\begin{aligned} |JM\Omega j_A k_A j_B k_B \Omega_B\rangle \\ = \mathbb{N}^{-1} D_{M\Omega}^{(J)*}(\alpha, \beta, 0) D_{\Omega_A k_A}^{(j_A)*}(\phi_A, \theta_A, \chi_A) \\ \times D_{\Omega_B k_B}^{(j_B)*}(\phi_B, \theta_B, \chi_B), \end{aligned} \quad (5)$$

where θ_F , ϕ_F and χ_F are the body-fixed Euler angles for fragment F , $\mathbb{N}$ is the normalizing factor for the three functions, M is the quantum number for projection of $\hat{J}$ onto the (spatially fixed) z -axis and Ω is the quantum number associated with the $\hat{J}_z$ operator for projection of $\hat{J}$ onto the (body-fixed) z -axis. Note that, by definition, $\Omega_A = \Omega - \Omega_B$. Since J and M are good quantum numbers, they will

henceforth be suppressed for notational simplicity. The basis set of Eq. (5) differs from that of Brocks et al. in that we do not make use of Clebsch–Gordan (C-G) coupling to obtain eigenfunctions of the $\hat{j}$ operator. As with their basis, evaluation of the action of $\hat{K}_A^{\text{ROT}}$ and $\hat{K}_B^{\text{ROT}}$ yields the standard form

$$\begin{aligned} \langle \Omega' j'_A k'_A j'_B k'_B \Omega'_B | \hat{K}_A^{\text{ROT}} + \hat{K}_B^{\text{ROT}} | \Omega j_A k_A j_B k_B \Omega_B \rangle \\ = \delta_{\Omega' \Omega} \delta_{j'_A j_A} \delta_{j'_B j_B} \delta_{\Omega'_B \Omega_B} (\delta_{k'_B k_B} h_{j_A k'_A k_A} + \delta_{k'_A k_A} h_{j_B k'_B k_B}), \end{aligned} \quad (6)$$

where

$$\begin{aligned} h_{j_F k'_F k_F} = \frac{1}{2} \delta_{k'_F k_F} [(B_F + C_F) j_F (j_F + 1) \\ + (2A_F - B_F - C_F) k_F^2] \\ + \frac{1}{4} \delta_{k'_F k_F \pm 2} (B_F - C_F) C_{j_F k_F}^{\pm} C_{j_F k_F \pm 1}^{\pm} \end{aligned} \quad (7)$$

and

$$C_{jk}^{\pm} = [j(j+1) - k(k \pm 1)]^{1/2}. \quad (8)$$

In Eq. (7), we have used the prolate top convention such that the A principal axis of a fragment is regarded as its fragment-fixed z -axis. It remains to evaluate the matrix elements of $\hat{K}_{\text{ORB}}$, Eq. (3). The actions of $\hat{j}^2$, $\hat{j}_{\pm}$, $\hat{j}_z$ and $\hat{j}_{\pm}$ on the basis functions of Eq. (5) are evaluated in a manner analogous to that which Brocks et al. used for their basis set [21], since both their basis and the present basis are (J, Ω) -resolved. The action of $\hat{j}^2$ and of $\hat{j}_{\pm}$ on the present basis set may be evaluated with the use of the following identities:

$$\hat{j}^2 = \hat{j}_A^2 + \hat{j}_B^2 + 2\hat{j}_z^A \hat{j}_z^B + \hat{j}_+^A \hat{j}_-^B + \hat{j}_-^A \hat{j}_+^B, \quad (9)$$

$$\hat{j}_{\pm} = \hat{j}_{\pm}^A + \hat{j}_{\pm}^B. \quad (10)$$

The matrix elements are then readily shown to be

$$\begin{aligned} \langle \Omega' j'_A k'_A j'_B k'_B \Omega'_B | \hat{K}_{\text{ORB}} | \Omega j_A k_A j_B k_B \Omega_B \rangle \\ = \delta_{j'_A j_A} \delta_{k'_A k_A} \delta_{j'_B j_B} \delta_{k'_B k_B} \frac{\hbar^2}{2\mu R^2} \\ \times \{ \delta_{\Omega' \Omega} \delta_{\Omega'_B \Omega_B} [J(J+1) - 2\Omega^2 + j_A(j_A+1) \\ + j_B(j_B+1) + 2\Omega_A \Omega_B] \\ + \delta_{\Omega' \Omega} \delta_{\Omega'_B \Omega_B \mp 1} C_{j_A \Omega_A}^{\pm} C_{j_B \Omega_B}^{\mp} - \delta_{\Omega' \Omega \pm 1} C_{j_A \Omega}^{\pm} \\ \times (\delta_{\Omega'_B \Omega_B} C_{j_A \Omega_A}^{\pm} + \delta_{\Omega'_B \Omega_B \pm 1} C_{j_B \Omega_B}^{\pm}) \} \end{aligned} \quad (11)$$

The rotational kinetic energy matrix summarized by Eqs. (6)–(8) and (11) is very sparse and hence may be stored, or generated as required.

Since in the study of reactions proceeding on a potential surface with no barrier to the recombination one must inevitably deal with large basis sets, our focus is to develop an algorithm for action of $\hat{H}_{\text{TM}}$ which functions efficiently within the context of iterative methods. Thus, one seeks to efficiently act with $\hat{H}_{\text{TM}}$ on a (typically large) state vector without having to store the full Hamiltonian matrix. With this consideration in mind, the action of the interaction potential is effected by a change of representation from the direct-product, body-fixed Wigner functions of Eq. (5) to a discrete variable representation (DVR) in the internal angular space of the system. In the DVR, the action of the potential on the state vector is evaluated simply by multiplication. For the general case of two nonlinear molecular fragments, $\hat{V}_{\text{INT}}$ depends on five of the body-fixed Euler angles of the two fragments. $\hat{V}_{\text{INT}}$ is independent of the external Euler angles α and β which orient the body-fixed z -axis, and, by symmetry of the rotation of the whole system about the z -axis, $\hat{V}_{\text{INT}}$ is effectively independent of one of the fragment ϕ angles. We shall assume that $\hat{V}_{\text{INT}}$ has been evaluated explicitly as a function of ϕ_B , i.e. $\hat{V}_{\text{INT}} = \hat{V}_{\text{INT}}(\theta_A, \chi_A, \phi_B, \theta_B, \chi_B)$. Thus, only the indexes $(j_A, k_A, j_B, k_B, \Omega_B)$ in the state vector need be transformed, so that in general a five-dimensional transformation is involved. The procedure for carrying out this transformation has been described by Leforestier [22]. After multiplication by the potential, back transformation to the Wigner basis then completes the operation of acting with $\hat{V}_{\text{INT}}$ on the state vector.

It is useful at this point to note some specific advantages of the present method. Firstly, within the context of a pseudo-spectral algorithm, the body-fixed Wigner-function representation of Eq. (5) has an advantage over the Clebsch–Gordan (C-G) coupled representation [21] in that the latter requires an additional transformation to ‘undo’ the C-G coupling before transforming to the angular grid. Consequently, there is an additional C-G coupling transformation required during the back-transformation from the grid. Both representations exhibit sparse kinetic energy matrices (of somewhat different structure), but the lower operations count for transformation of

the state vector to and from the DVR using the present Wigner representation suggests that it may prove to be more efficient than the C-G coupled representation, since it is the transformation to and from the DVR which occupies a major part of the computing time in multidimensional applications where very large basis sets are involved. Secondly, the representation of this work also facilitates simplified evaluation of potential matrix elements using vector coupling algebra [23] for cases where the spectral bandwidth of the potential is sufficiently small that this approach is favourable to (i.e. numerically cheaper than) transforming to the angular grid. This is expected to be the case, for example, at sufficiently large separations between the molecular fragments where a low-order multipole expansion of the potential suffices. Expressions for these matrix elements analogous to those provided by Brocks et al. for the C-G coupled representation [21] will be provided in a forthcoming paper [24]. Finally, the present pseudo-spectral approach facilitates computations for non-zero total angular momentum. Since the basis vectors are eigenfunctions of the total angular momentum operator, an initial seed vector for the Lanczos algorithm is readily selected which contains components only of eigenvectors of the Hamiltonian with a given total angular momentum. This is a much more difficult (and as yet unresolved) problem for a purely DVR-based computation. Calculations for non-zero total angular momentum are, however, essential in order to treat reaction kinetics on potential surfaces with no barrier to recombination [25].

3. Calculations

We have employed the Lanczos algorithm [26] to calculate eigenenergies and eigenfunctions for the transitional modes of ketene during its dissociation via a loose transition state on the ground singlet electronic surface. This corresponds to the case of an asymmetric top plus a linear rotor, for which the body-fixed basis set and kinetic energy matrix elements are obtained as a special case of the formulae of the previous section. For the present illustrative purposes, we have chosen the analytic model for the interaction potential between the CH_2 and CO frag-

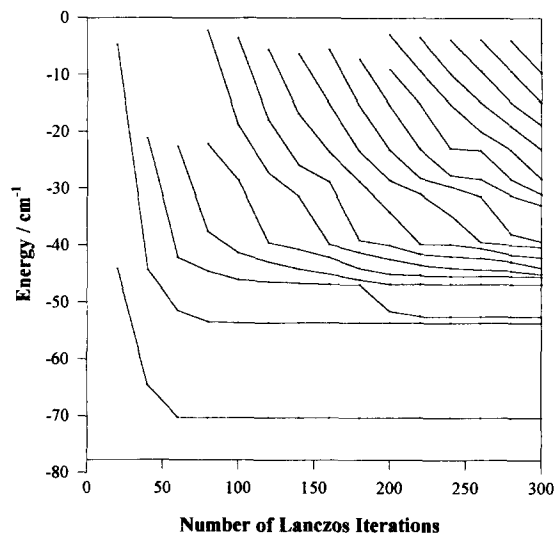


Fig. 1. Plot of negative eigenvalues obtained from the Lanczos algorithm as a function of the number of iterations, calculated at a center-of-mass separation of 4.0 Å. Data are plotted for numbers of iterations increasing incrementally by 25.

ments used by Klippenstein and Marcus [7]. The interaction potential is written as the sum of an effective bonding potential and non-bonding interactions,

$$V_{\text{int}} = V_{\text{b}} + V_{\text{nb}}, \quad (12)$$

where the bonding potential is given by an effective Varshni potential, modified by simple angular anisotropy factors [7], and the non-bonding potential is described by two-body Lennard-Jones interactions between all pairs of atoms except the carbon–carbon pair.

Our calculations have been carried out for zero total angular momentum, hence $J = M = \Omega = 0$. The upper limits on the remaining indices of our Wigner basis functions were set at $j_{\text{max}}(\text{CH}_2) = 15$, $k_{\text{max}}(\text{CH}_2) = 10$, $j_{\text{max}}(\text{CO}) = 25$, $\Omega_{\text{max}}(\text{CO}) = 10$, leading to a basis of $\approx 2 \times 10^5$ functions. These limits were arrived at by testing for convergence of the lowest energy state at the smallest center-of-mass separation considered, 3.0 Å.

Figs. 1 and 2 present results obtained for a center-of-mass separation of 4.0 Å. In Fig. 1, the eigenvalues obtained with increasing numbers of Lanczos iterations are plotted. At this separation the minimum interaction potential relative to the zero potential of

the infinitely separated product fragments is -102 cm^{-1} , and for any given number of iterations we have plotted only those eigenvalues obtained which lie below zero. As is clearly evident, the smallest eigenvalues converge the most quickly, and convergence becomes increasingly more difficult to obtain as the energy increases and the spectrum becomes more dense. Such behaviour is typical of the Lanczos algorithm, which generally converges the sparse regions of the spectrum most quickly (e.g. Ref. [26]). Fig. 2 shows the rate of convergence of the eigenvectors corresponding to the five eigenvalues lowest in energy, expressed in terms of the energy dispersion which is conveniently evaluated as the norm of the residual vector

$$\sigma_j = \langle \psi_j | (H_{\text{TM}} - \bar{E})^2 | \psi_j \rangle^{1/2}. \quad (13)$$

In Eq. (13), $\bar{E}$ is the expectation value associated with the approximate eigenvector ψ_j ,

$$\bar{E} = \langle \psi_j | H_{\text{TM}} | \psi_j \rangle \quad (14)$$

Comparison of Figs. 1 and 2 illustrates that the eigenvalues converge more quickly than the associated eigenvectors, and that the eigenvectors associ-

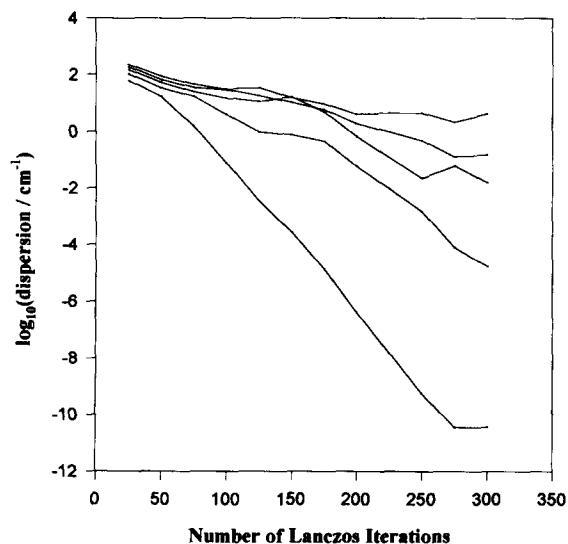


Fig. 2. Plot of the log of the dispersion associated with eigenvectors as a function of the number of Lanczos iterations. The lines correspond to the eigenvectors associated with the lowest five eigenvalues at center-of-mass separation 4.0 Å .

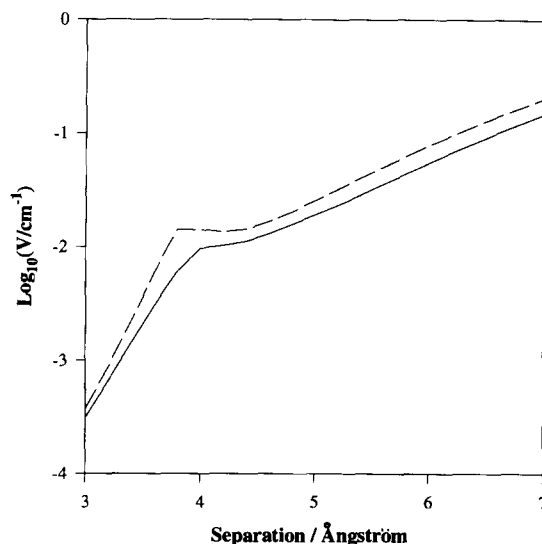


Fig. 3. Plot of the log of the lowest eigenvalue (i.e., the zero-point energy) for the transitional modes of ketene as a function of center-of-mass separation (dashed line) compared with the minimum path on the potential surface (full line).

ated with the lowest eigenenergies are the first to converge.

Figs. 3–5 illustrate different aspects of the lowest-energy eigenstate of the transitional modes for center-of-mass separations ranging from 3.0 to 7.0 Å . In Fig. 3 is plotted the zero-point energy (dashed line) in comparison with the minimum energy pathway on the potential surface for the chosen reaction coordinate. Figs. 4 and 5 provide information about the nature of the lowest-energy eigenfunction at different separations. The cumulative norm of the multidimensional eigenfunction is plotted as a function of the j quantum number for methylene in Fig. 4, and as a function of the j quantum number for carbon monoxide in Fig. 5. The lowest energy state clearly corresponds to j quantum numbers of zero for both fragments at large separations since this is the lowest energy state of the isolated fragments. At the smallest separation, 3.0 Å , the lowest energy state contains components from a range of j quanta of both fragments. This range is greater for CO than for CH_2 , as might be expected on the basis of intuitive reasoning that the smaller free-rotor state spacing of CO allows its free-rotor states to be more easily coupled by the potential. In relation to the

maximum bounds set for the basis set, we note that even though the cumulative norm as a function of the j^{CO} quantum number appears to become essentially zero when j^{CO} is 15 (the maximum in Fig. 5), a maximum value of 25 was necessary to converge the associated eigenvalue (the zero-point energy for the transitional modes) to within 1 cm^{-1} . Setting $j_{\text{max}}^{\text{CO}} = 15$ resulted in an eigenvalue in error by 30 cm^{-1} . The fact that surprisingly large basis sets are required in order to converge accurately rovibrational states has been noted in a different context by Leforestier and Iung [27].

It is noteworthy in addition that the lowest-energy eigenfunction attains its ' $j = 0$ ' character within that part of the angular coordinate space associated with CH_2 when the separation is about $3.8 \text{ \AA}$, whereas in that part of the space associated with CO, higher j quanta are mixed in until about $6.6 \text{ \AA}$. This observation has interesting implications with regards a possi-

ble contrast in the extent of adiabatic behavior of the two fragments during the dissociation process.

4. Conclusions

We have presented a pseudo-spectral algorithm for the action of the transitional mode Hamiltonian on a state vector in a body-fixed Wigner function representation. This algorithm is ideally suited for implementation within iterative algorithms and is capable of operating efficiently with very large basis sets, as is typically required for reactions proceeding on a potential surface with no barrier to the recombination process. Illustrative calculations have been carried out using the Lanczos algorithm in order to compute accurate, converged eigenenergies and eigenfunctions for the hindered rotational degrees of freedom of the methylene and carbon monoxide

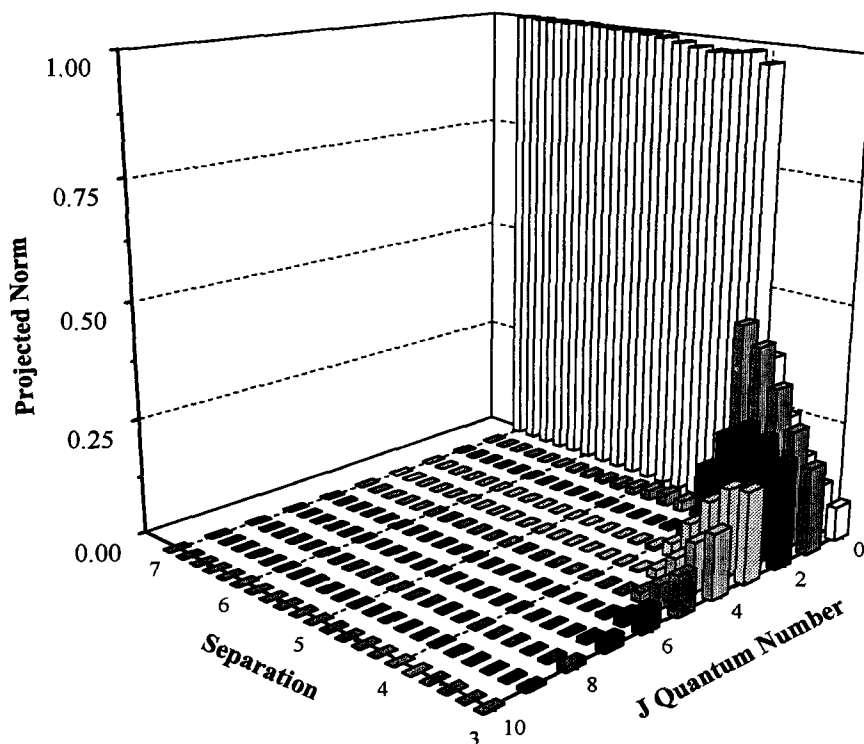


Fig. 4. Plot of the cumulative norm of the lowest energy eigenfunction expressed as a function of the j rotational quantum number of CH_2 and as a function of the center-of-mass separation ($\text{\AA}$).

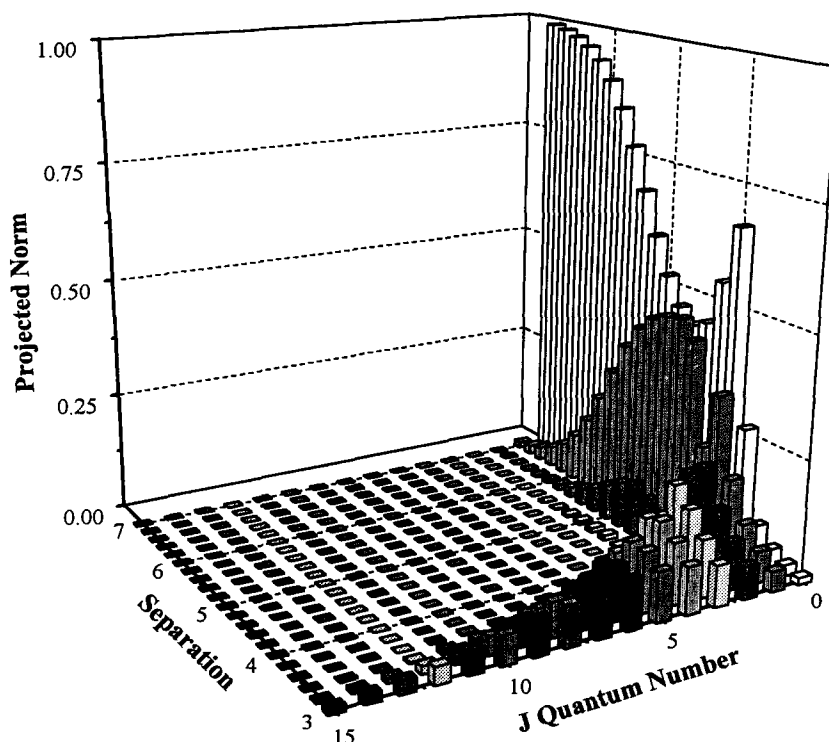


Fig. 5. Plot of the cumulative norm of the lowest energy eigenfunction expressed as a function of the j rotational quantum number of CO and as a function of the center-of-mass separation (Å).

fragments in the dissociation of ketene on the ground singlet electronic surface. The Lanczos algorithm converges the lowest energy states first, with states higher in energy becoming successively more difficult to obtain.

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References

- [1] C.X.W. Qian, M. Noble, I. Nadler, H. Reisler and C. Wittig, *J. Chem. Phys.* 83 (1985) 5573.
- [2] C. Wittig, I. Nadler, H. Reisler, M. Noble, J. Catanzarite and G. Radhakrishnan, *J. Chem. Phys.* 83 (1985) 5581.
- [3] I.C. Chen, W.H. Green Jr. and C.B. Moore, *J. Chem. Phys.* 89 (1988) 314.
- [4] W.H. Green Jr., I.C. Chen and C.B. Moore, *Ber. Bunsenges. Physik. Chem.* 92 (1988) 389.
- [5] R.A. Marcus, *Chem. Phys. Letters* 144 (1988) 208.
- [6] S.J. Klippenstein, L.R. Khundkar, A.H. Zewail and R.A. Marcus, *J. Chem. Phys.* 89 (1988) 4761.
- [7] S.J. Klippenstein and R.A. Marcus, *J. Chem. Phys.* 91 (1989) 2280.
- [8] D.C. Robie, M. Hunter, J.L. Bates and H. Reisler, *Chem. Phys. Letters* 193 (1992) 413.
- [9] U. Peskin, W.H. Miller and H. Reisler, *J. Chem. Phys.* (1995) preprint.
- [10] I. Garcia-Moreno, E.R. Lovejoy and C.B. Moore, *J. Chem. Phys.* 100 (1994) 8890.
- [11] D.M. Wardlaw and R.A. Marcus, *Advan. Chem. Phys.* 70 (1988) 231.
- [12] M. Quack and J. Troe, *Ber. Bunsenges. Physik. Chem.* 78 (1974) 240.
- [13] E.E. Nikitin, J. Troe and V.G. Ushakov, *J. Chem. Phys.* 102 (1995), 4101.
- [14] D.C. Clary, *Ann. Rev. Phys. Chem.* 41 (1990) 61.
- [15] J.M. Hutson and A.E. Thornley, *J. Chem. Phys.* 100 (1994) 2505.

- [16] M.-L. Dubernet and J.M. Hutson, *J. Chem. Phys.* 101 (1994) 1939.
- [17] M.-L. Dubernet and R. McCarroll, *Z. Physik D* 13 (1989) 225; 15 (1990) 333.
- [18] E.E. Nikitin and J. Troe, *J. Chem. Phys.* 92 (1990) 6594.
- [19] A.I. Maergoiz, E.E. Nikitin and J. Troe, *J. Chem. Phys.* 95 (1991) 5117.
- [20] M. Quack and M.A. Suhm, *J. Chem. Phys.* 95 (1991) 28.
- [21] G. Brocks, A. van der Avoird, B.T. Sutcliffe and J. Tennyson, *Mol. Phys.* 50 (1983) 1025.
- [22] C. Leforestier, *J. Chem. Phys.* 101 (1994) 7357.
- [23] R.N. Zare, *Angular momentum* (Wiley, New York, 1988).
- [24] S.C. Smith, in preparation.
- [25] R.G. Gilbert and S.C. Smith, *Theory of unimolecular and recombination reactions* (Blackwell, Oxford, 1990).
- [26] J.K. Cullum and R.A. Willoughby, *Lanczos algorithms for large symmetric eigenvalue computations, Vol. I* (Birkhäuser, Boston, 1985).
- [27] C. Iung and C. Leforestier, *J. Chem. Phys.* 97 (1992) 2481.